

EMCH 792: Optimal State Estimation
Homework 4
Written Portion Solutions

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Exercise 5.1: Radioactive Mass Kalman Filter

A radioactive mass has a half-life of τ seconds. At each time step the number of emitted particles x is half of what it was one time step ago, but there is some error u_k (zero-mean with variance Q) in the number of emitted particles due to background radiation. At each time step, the number of emitted particles is counted. The instrument used to count the number of emitted particles has a random error at time k of v_k , which is zero-mean with a variance of R . Assume that u_k and v_k are uncorrelated.

Derivation

(a) Write the linear system equations for this system.

This is a scalar system. We assume the discrete sampling period equals one half-life interval ($\Delta t = \tau$), so the number of emitted particles halves at each time step. Including the process noise u_k (which we denote w_k to follow standard Kalman filter notation) and the measurement noise v_k , the linear system equations are

$$x_{k+1} = \frac{1}{2} x_k + w_k, \quad w_k \sim (0, Q) \quad (1)$$

$$y_k = x_k + v_k, \quad v_k \sim (0, R) \quad (2)$$

The system matrices are therefore $F = 1/2$, $H = 1$, and there is no deterministic control input ($G = 0$). The process and measurement noises are uncorrelated, i.e., $E[w_k v_j^T] = 0$ for all k, j .

Derivation

(b) Suppose we want to use a Kalman filter to find the optimal estimate of the number of emitted particles at each time step. Write the one-step *a posteriori* Kalman filter equations for this system.

Applying the standard discrete-time Kalman filter (Simon, Eq. 5.27) to the scalar system from part (a), the one-step *a posteriori* filter equations are as follows.

Time update (prediction):

$$\hat{x}_k^- = \frac{1}{2} \hat{x}_{k-1}^+ \quad (3)$$

$$P_k^- = \frac{1}{4} P_{k-1}^+ + Q \quad (4)$$

Measurement update (correction):

$$K_k = \frac{P_k^-}{P_k^- + R} \quad (5)$$

$$\hat{x}_k^+ = \hat{x}_k^- + K_k (y_k - \hat{x}_k^-) = (1 - K_k) \frac{1}{2} \hat{x}_{k-1}^+ + K_k y_k \quad (6)$$

$$P_k^+ = (1 - K_k) P_k^- \quad (7)$$

Note that since $H = 1$, the innovation is simply $y_k - \hat{x}_k^-$, and the gain simplifies to the ratio of the prior variance to the total variance.

Derivation

(c) Find the steady-state *a posteriori* estimation-error variance for the Kalman filter.

At steady state, $P_k^- \rightarrow P^-$ and $P_k^+ \rightarrow P^+$, so the Riccati recursion becomes time-invariant. From Eq. (7) and (5),

$$P^+ = (1 - K) P^- = \frac{P^- R}{P^- + R} \quad (8)$$

Substituting into the prediction equation (4),

$$P^- = \frac{1}{4} P^+ + Q = \frac{1}{4} \cdot \frac{P^- R}{P^- + R} + Q \quad (9)$$

Multiplying both sides by $(P^- + R)$,

$$P^- (P^- + R) = \frac{1}{4} P^- R + Q(P^- + R) \quad (10)$$

$$(P^-)^2 + P^- R = \frac{1}{4} P^- R + Q P^- + Q R \quad (11)$$

$$(P^-)^2 + \frac{3}{4} R P^- - Q P^- - Q R = 0 \quad (12)$$

$$(P^-)^2 + \left(\frac{3R}{4} - Q\right) P^- - Q R = 0 \quad (13)$$

Applying the quadratic formula and taking the positive root,

$$P^- = \frac{-(3R/4 - Q) + \sqrt{(3R/4 - Q)^2 + 4QR}}{2} \quad (14)$$

The steady-state *a posteriori* variance is then

$$P^+ = \frac{P^- R}{P^- + R} \quad (15)$$

Substituting Eq. (14) into Eq. (15) gives the explicit closed form

$$P^+ = \frac{R \left(-(3R/4 - Q) + \sqrt{(3R/4 - Q)^2 + 4QR} \right)}{2R - (3R/4 - Q) + \sqrt{(3R/4 - Q)^2 + 4QR}} \quad (16)$$

Derivation

(d) What is the steady-state Kalman gain when $Q = R$? What is the steady-state Kalman gain when $Q = 2R$? Give an intuitive explanation for why the steady-state gain changes the way it does when the ratio of Q to R changes.

Case 1: $Q = R$. Substituting into Eq. (13),

$$(P^-)^2 + \left(\frac{3R}{4} - R\right) P^- - R^2 = 0 \implies (P^-)^2 - \frac{R}{4} P^- - R^2 = 0 \quad (17)$$

Using the quadratic formula,

$$P^- = \frac{R/4 + \sqrt{R^2/16 + 4R^2}}{2} = \frac{R/4 + \sqrt{65R^2/16}}{2} = \frac{(1 + \sqrt{65})R}{8} \approx 1.133R \quad (18)$$

The steady-state Kalman gain is

$$K = \frac{P^-}{P^- + R} = \frac{1 + \sqrt{65}}{9 + \sqrt{65}} \approx \boxed{0.531} \quad (19)$$

Case 2: $Q = 2R$. Substituting into Eq. (13),

$$(P^-)^2 + \left(\frac{3R}{4} - 2R\right) P^- - 2R^2 = 0 \implies (P^-)^2 - \frac{5R}{4} P^- - 2R^2 = 0 \quad (20)$$

Using the quadratic formula,

$$P^- = \frac{5R/4 + \sqrt{25R^2/16 + 8R^2}}{2} = \frac{5R/4 + \sqrt{153R^2/16}}{2} = \frac{(5 + \sqrt{153})R}{8} \approx 2.171R \quad (21)$$

The steady-state Kalman gain is

$$K = \frac{P^-}{P^- + R} = \frac{5 + \sqrt{153}}{13 + \sqrt{153}} \approx \boxed{0.685} \quad (22)$$

Intuition. When Q/R increases, the process model becomes less trustworthy relative to the measurements (the system dynamics are noisier while measurement quality stays the same). The Kalman filter compensates by increasing its gain K , placing more weight on the measurements and less weight on the model prediction. This is consistent with the general principle: a higher Kalman gain means more trust in the measurements, which is appropriate when process noise dominates measurement noise.

Final Result

Exercise 5.1 Summary. For the radioactive mass system with $F = 1/2$, $H = 1$:

- *A posteriori* Kalman filter: $\hat{x}_k^+ = (1 - K_k)\frac{1}{2}\hat{x}_{k-1}^+ + K_k y_k$, $K_k = P_k^- / (P_k^- + R)$
- Steady-state prior variance: $(P^-)^2 + (3R/4 - Q)P^- - QR = 0$ (positive root)
- Steady-state gain ($Q = R$): $K \approx 0.531$
- Steady-state gain ($Q = 2R$): $K \approx 0.685$

Exercise 5.4: Fish Tank Kalman Filter

Suppose that you have a fish tank with x_p piranhas and x_g guppies [Bay99]. Once per week, you put guppy food into the tank (which the piranhas do not eat). Each week the piranhas eat some of the guppies. The birth rate of the piranhas is proportional to the guppy population, and the death rate of the piranhas is proportional to their own population (due to overcrowding). Therefore $x_p(k+1) = x_p(k) + k_1 x_g(k) - k_2 x_p(k) + u_p(k)$, where k_1 and k_2 are proportionality constants and $u_p(k)$ is white noise with a variance of one that accounts for mismodeling. The birth rate of the guppies is proportional to the food supply u , and the death rate of the guppies is proportional to the piranha population. Therefore, $x_g(k+1) = x_g(k) + u(k) - k_3 x_p(k) + u_g(k)$, where k_3 is a proportionality constant and $u_g(k)$ is white noise with a variance of one that accounts for mismodeling. The step size for this model is one week. Every week, you count the piranhas and guppies. You can count the piranhas accurately because they are so large, but your guppy count has zero-mean noise with a variance of one. Assume that $k_1 = 1$ and $k_2 = k_3 = 1/2$.

Derivation

(a) Generate a linear state-space model for this system.

Define the state vector $\mathbf{x}_k = \begin{bmatrix} x_p(k) \\ x_g(k) \end{bmatrix}$. Using the given difference equations with $k_1 = 1$ and $k_2 = k_3 = 1/2$,

$$x_p(k+1) = x_p(k) + k_1 x_g(k) - k_2 x_p(k) + u_p(k) = \frac{1}{2}x_p(k) + x_g(k) + u_p(k) \quad (23)$$

$$x_g(k+1) = x_g(k) + u(k) - k_3 x_p(k) + u_g(k) = -\frac{1}{2}x_p(k) + x_g(k) + u(k) + u_g(k) \quad (24)$$

In matrix form, the state equation is

$$\mathbf{x}_{k+1} = F \mathbf{x}_k + G u_k + \mathbf{w}_k \quad (25)$$

where

$$F = \begin{bmatrix} 1/2 & 1 \\ -1/2 & 1 \end{bmatrix}, \quad G = \begin{bmatrix} 0 \\ 1 \end{bmatrix}, \quad Q = E[\mathbf{w}_k \mathbf{w}_k^T] = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = I_2 \quad (26)$$

The piranhas are counted perfectly, and the guppy count has zero-mean noise with variance 1. The measurement equation is

$$\mathbf{y}_k = H \mathbf{x}_k + \mathbf{v}_k \quad (27)$$

where

$$H = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = I_2, \quad R = \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} \quad (28)$$

Derivation

(b) Suppose that at the initial time you have a perfect count for x_p and x_g . Using a Kalman filter to estimate the guppy population, what is the variance of your guppy population estimate after one week? What is the variance after two weeks?

A perfect initial count means $P_0^+ = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$.

Week 1. Time update:

$$P_1^- = F P_0^+ F^T + Q = \begin{bmatrix} 1/2 & 1 \\ -1/2 & 1 \end{bmatrix} \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} 1/2 & -1/2 \\ 1 & 1 \end{bmatrix} + I_2 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \quad (29)$$

Kalman gain:

$$K_1 = P_1^- H^T (H P_1^- H^T + R)^{-1} = I_2 \left(I_2 + \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} \right)^{-1} = \begin{bmatrix} 1 & 0 \\ 0 & 1/2 \end{bmatrix} \quad (30)$$

Measurement update (Joseph form, used for numerical robustness with singular R):

$$\begin{aligned} P_1^+ &= (I_2 - K_1) P_1^- (I_2 - K_1)^T + K_1 R K_1^T \\ &= \begin{bmatrix} 0 & 0 \\ 0 & 1/2 \end{bmatrix} \end{aligned} \quad (31)$$

The guppy variance after one week is $P_1^+(2, 2) = \boxed{1/2}$.

Week 2. Time update:

$$\begin{aligned} P_2^- &= F P_1^+ F^T + Q = \begin{bmatrix} 1/2 & 1 \\ -1/2 & 1 \end{bmatrix} \begin{bmatrix} 0 & 0 \\ 0 & 1/2 \end{bmatrix} \begin{bmatrix} 1/2 & -1/2 \\ 1 & 1 \end{bmatrix} + I_2 \\ &= \begin{bmatrix} 0 & 1/2 \\ 0 & 1/2 \end{bmatrix} \begin{bmatrix} 1/2 & -1/2 \\ 1 & 1 \end{bmatrix} + I_2 = \begin{bmatrix} 1/2 & 1/2 \\ 1/2 & 1/2 \end{bmatrix} + I_2 = \begin{bmatrix} 3/2 & 1/2 \\ 1/2 & 3/2 \end{bmatrix} \end{aligned} \quad (32)$$

Kalman gain:

$$H P_2^- H^T + R = \begin{bmatrix} 3/2 & 1/2 \\ 1/2 & 3/2 \end{bmatrix} + \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 3/2 & 1/2 \\ 1/2 & 5/2 \end{bmatrix} \quad (33)$$

The inverse is $\frac{1}{3/2 \cdot 5/2 - 1/4} \begin{bmatrix} 5/2 & -1/2 \\ -1/2 & 3/2 \end{bmatrix} = \frac{1}{7/2} \begin{bmatrix} 5/2 & -1/2 \\ -1/2 & 3/2 \end{bmatrix} = \begin{bmatrix} 5/7 & -1/7 \\ -1/7 & 3/7 \end{bmatrix}$.

$$K_2 = P_2^- (H P_2^- H^T + R)^{-1} = \begin{bmatrix} 3/2 & 1/2 \\ 1/2 & 3/2 \end{bmatrix} \begin{bmatrix} 5/7 & -1/7 \\ -1/7 & 3/7 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 1/7 & 4/7 \end{bmatrix} \quad (34)$$

Measurement update (Joseph form):

$$\begin{aligned} P_2^+ &= (I_2 - K_2) P_2^- (I_2 - K_2)^T + K_2 R K_2^T \\ &= \begin{bmatrix} 0 & 0 \\ 0 & 4/7 \end{bmatrix} \end{aligned} \quad (35)$$

The guppy variance after two weeks is $P_2^+(2, 2) = \boxed{4/7}$.

Derivation

(c) What is the ratio of the piranha population to the guppy population when they reach steady state? Assume that the process noise is zero for this part of the problem.

At steady state, $\mathbf{x}_{k+1} = \mathbf{x}_k = \mathbf{x}_{ss}$ and the process noise is zero. From the state equation,

$$\mathbf{x}_{ss} = F \mathbf{x}_{ss} + G u_{ss} \implies (I - F) \mathbf{x}_{ss} = G u_{ss} \quad (36)$$

Computing $I - F$,

$$(I - F) = \begin{bmatrix} 1/2 & -1 \\ 1/2 & 0 \end{bmatrix} \quad (37)$$

The system becomes

$$\begin{bmatrix} 1/2 & -1 \\ 1/2 & 0 \end{bmatrix} \begin{bmatrix} x_{p,ss} \\ x_{g,ss} \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \end{bmatrix} u_{ss} \quad (38)$$

From the second row: $\frac{1}{2}x_{p,ss} = u_{ss}$, so $x_{p,ss} = 2u_{ss}$.

From the first row: $\frac{1}{2}x_{p,ss} - x_{g,ss} = 0$, so $x_{g,ss} = \frac{1}{2}x_{p,ss} = u_{ss}$.

Therefore, the steady-state ratio of piranhas to guppies is

$$\boxed{\frac{x_{p,ss}}{x_{g,ss}} = \frac{2u_{ss}}{u_{ss}} = 2} \quad (39)$$

At steady state, there are twice as many piranhas as guppies.

Final Result

Exercise 5.4 Summary. For the fish tank system with state $\mathbf{x} = [x_p; x_g]^T$:

- State-space model: $F = \begin{bmatrix} 1/2 & 1 \\ -1/2 & 1 \end{bmatrix}$, $G = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$, $H = I_2$, $Q = I_2$, $R = \text{diag}(0, 1)$
- Guppy variance after 1 week: $P_1^+(2, 2) = 1/2$
- Guppy variance after 2 weeks: $P_2^+(2, 2) = 4/7$
- Steady-state population ratio: $x_p/x_g = 2$